

CHRISANN FERRAO

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SUMMARY

Electronics Engineering graduate with 1+ year of hands-on experience building computer vision and edge-AI systems for industrial automation. Designed and deployed multiple automated inspection systems at Godrej Enterprises Group that delivered an 800% increase in inspection throughput and 75% reduction in manual inspection time. Skilled across the full development stack — from hardware integration and embedded firmware to real-time image processing, ML model experimentation, and web-based monitoring dashboards.

EXPERIENCE

Automation Engineer (Trainee) — Godrej Enterprises Group

Aug 2024 – Present

Madkai, Goa

E-Locks Assembly Line Automation

- Built a full-stack web application (HTML/CSS, JavaScript, Python Flask) for real-time assembly line monitoring and inspection management.
- Developed an automated inspection system on Raspberry Pi 5 using a Logitech BRIO webcam for lock capacitive-sense pattern verification and fingerprint LED colour validation.
- Integrated a SICK photoelectric proximity sensor for deadbolt latch detection and a Sunfound USB microphone for audio testing of lock speakers.
- Researched neural network approaches (TensorFlow, PyTorch) for audio classification on ~50 labelled samples; evaluated CNNs and lightweight RNN architectures before finalising MFCC feature extraction with Cosine Similarity for production reliability.
- Implemented speech verification using the Vosk speech recognition model to detect predefined verbal phrases.
- Increased inspection throughput by 800%, scaling from ~500 to 4,500 units/month through automated real-time quality verification.

Key Number Detection System (OCR)

- Developed a computer vision OCR system for automatic detection and validation of engraved key numbers using a Raspberry Pi HQ Camera with a 16 mm lens.
- Designed an image preprocessing pipeline (CLAHE, noise reduction, binary enhancement) for robust performance under variable lighting conditions.
- Reduced manual key identification time by 70%; deployed as a lightweight edge solution on Raspberry Pi hardware.

Key Ward Cutting Inspection System

- Built a high-precision vision measurement system for lever key ward dimensions, achieving sub-40 μm accuracy and better than 5 μm repeatability.
- Implemented on Raspberry Pi 5 with a VS Technology telecentric lens and collimated backlight for distortion-minimised silhouette capture and reliable edge extraction.
- Applied sub-pixel edge detection, curve fitting, and RANSAC-based calibration for accurate reconstruction of complex key geometries.
- Reduced manual inspection time by 75% by automating dimensional verification and real-time deviation analysis against engineering drawings.

Intern — UR Rao Satellite Centre (ISRO)

Jul 2024 – Aug 2024

Bengaluru, Karnataka

- Developed an audio data acquisition and processing subsystem for the Gaganyaan mission using a microphone, ADC, and FPGA (VHDL) for data storage and playback.
- Wrote VHDL testbenches to validate storage behaviour and verified real-world functionality via UART communication between the FPGA and PC, visualising data with a C program.

PROJECTS

Navigation & Localization of Unmanned Underwater Vehicles

Sept 2024 – May 2025

- Developed vision-based 6DOF localization using AprilTags as a low-cost alternative to acoustic positioning; benchmarked AprilTag vs. ArUco, selecting AprilTag for superior robustness and partial-occlusion handling.

- Implemented two algorithms: single-camera PnP pose estimation per tag, and a dual-camera PnP/P3P/EPnP setup with calibrated intrinsics for faster, more precise localization.
- Migrated stack from Raspberry Pi 3B to Raspberry Pi 4 (BlueOS); streamed pose data to a Pixhawk flight controller with vision/IMU fusion for motor control and error correction.

ML Experiments — Audio Classification & MNIST

2024 – 2025

- Explored deep learning approaches for small-sample audio classification using TensorFlow and PyTorch; evaluated CNNs and lightweight RNN architectures across ~50 labelled samples before settling on a classical pipeline for production deployment.
- Conducted research experiments on MNIST to study feature extraction, model generalisation, and classification performance across standard and modified neural network architectures.

Clustering using Genetic Algorithm

Sept – Oct 2024

- Implemented a genetic algorithm for data point clustering, optimising centroid positions through selection, crossover, and mutation; benchmarked against K-Means across 100 data points.
- Demonstrated rapid convergence in as few as 1–2 generations using a GA pipeline of 50 candidates, 3 centroids, and 20–30 generations; visualised evolution dynamically with Matplotlib.

Arduino Breakout Circuit Board — PCB Design

Mar – Apr 2024

- Designed a custom Arduino-based breakout PCB in KiCad for teaching robotics to beginners, integrating multiple sensors into a single plug-and-play board to eliminate wiring complexity.
- Programmed and tested the board end-to-end, enabling hands-on engagement with core robotics concepts with minimal setup.

Document Scanner

Oct – Nov 2023

- Built a Python/OpenCV document scanner with a full pipeline: Canny edge detection, contour extraction, perspective transformation, and image enhancement — achieving reliable detection on skewed documents.

Airplane Detection and Radio

Mar – Apr 2023

- Explored airplane detection and FM radio reception using PlutoSDR and MATLAB, experimenting with RF signal processing and real-time signal demodulation.

TECHNOLOGIES

Languages: Python, C, C++, JavaScript, HTML/CSS, Arduino, VHDL, Verilog, R

Computer Vision & Image Processing: OpenCV, EasyOCR, CLAHE, MFCC, Canny Edge Detection, Sub-pixel Edge Detection, RANSAC, Perspective Transformation, Histogram Equalization

Machine Learning & Deep Learning: TensorFlow, PyTorch, CNNs, Genetic Algorithms, K-Means Clustering, Cosine Similarity, Vosk Speech Recognition

Edge AI & Embedded Systems: Raspberry Pi, FPGA, KiCad, PCB Design, UART, SICK Sensors, Edge-AI Deployment

Frameworks & Tools: Flask, NumPy, Matplotlib, MATLAB, PlutoSDR

Signal Processing: MFCC, SDR, RF Signal Processing, VLSI, Embedded Systems

EDUCATION

B.E. Electronics and Telecommunication

2021 – 2025

Goa College of Engineering, Farmagudi, Goa

CGPA: 8.69

Relevant Coursework: Computer Networks, VLSI, Control Systems, Signal Processing, AI / ML, Embedded Systems, Communication Systems

HSSCE (Science)

2021

Don Bosco Higher Secondary School, Panjim, Goa

86.33%

SSCE

2019

Sharada English High School, Marcel, Goa

91.33%

ACHIEVEMENTS & AWARDS

- **1st Place — Technical Treasure Hunt:** College Fest (Technix), 2025 — Puzzle-solving and technical aptitude competition
- **1st Place — Escape Room:** College Fest (Technix), 2025 — Logic, lateral thinking, and team problem-solving

- **IEEEExtreme 18.0:** IEEE, 2024 — 24-hour global competitive programming hackathon

CERTIFICATIONS

- Data Science for Engineers — NPTEL
- Python for Data Science — NPTEL
- Computer Graphics — NPTEL
- Verilog HDL and FPGA Design Flow — Goa College of Engineering
- CAD Modelling and 3D Printing — Goa College of Engineering

VOLUNTEER EXPERIENCE

Chairperson — IEEE Student Branch — WIE Council

- Led the organization of a South India-wide AI Summit featuring a Women's Summit with distinguished speakers from ISRO, AI industry, and academia, coordinating all logistics including lodging, scheduling, and speaker arrangements for 1,000+ participants.
- Orchestrated multiple technical competitions including coding, PCB design, and robotics as part of the core organizing team, driving cross-disciplinary engagement across South India.
- Designed and developed an online technical treasure hunt for Women's Day celebration, integrating clues based on prominent female figures with technical aptitude challenges, achieving 100+ participants with zero resource expenditure.
- Organized and volunteered for over 10 workshops and competitions for college and school students, consistently delivering strong participation rates and positive financial outcomes.

INTERESTS

Keyboard (10+ years) and vocals — perform regularly at college events and informal settings. Enjoys competitive puzzle-solving, escape rooms, and logic-based challenges.